The Boundary is a Rate: Coherence–Rupture–Regeneration and the Unevidenceable Boundary

Revised in the current CRR form, October 2026 (replaces the June 2026 edition)

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Abstract.

- **The argument I respond to.** Sandved-Smith, Fields, Doctor, Laukkonen and Hohwy argue that a persisting, self-evidencing system can never obtain evidence that it is separate from its environment. On their reading, realising emptiness is the embodied recognition of that impossibility.
- **What the June 2026 edition did, and why it is replaced.** The June edition read their boundary as a *rate* rather than a surface. It supported this with an older form of Coherence–Rupture–Regeneration (CRR), several parts of which have since been withdrawn:
  - the cut as Cramér–Rao saturation, $C \cdot \Omega = 1$;
  - separate thresholds for bistable and rotational systems, with a "topological" ratio of 2;
  - a parameter-free breathing prediction;
  - an amplifying memory kernel.
- **What this edition does.** It rebuilds the formal part in the current form, where the cut is a first passage of net Fisher–Rao travel of a half-turn and regeneration is an age-weighted Fréchet mean. It reports four checks, preregistered and confirmed on fresh random instances (verdict RESULT; ceiling "PASS-0; weak anchor").
  1. **Self-measurement overshoot.** A finite agent's measured boundary entropy overshoots the true value in every basis except the eigenbasis. This is the Schur–Horn theorem, so it is a theorem, not merely an illustration.
  2. **Retensing as continuity.** The June "retensing bound", that accumulated coherence C bounds the forbidden entanglement entropy S_A, is the continuity of entropy along a Fisher–Rao path, with the root varentropy as its constant. The constant is below 1 only for small systems. So $C \geq S_A$ holds for a qubit, but fails for larger systems: on every seed at $d = 32$ and, by an exact geodesic construction, at $d \geq 14$.
  3. **Regeneration.** It is a many-to-one, normalised average: neither amplifying nor invertible.
  4. **The collective now.** Coupled CRR rotors are the Kuramoto model and reproduce its transition.
- **What survives.** The ontological reading: the boundary as a rate, a present with no content of its own, and a far side that is the settled past of other finite systems. It is restated as a philosophical position, not a result.

Keywords: self-evidencing; emptiness; quantum reference frames; Fisher information; varentropy; Schur–Horn; synchronisation; process ontology; epistemic check.

1. What changed since June

June 2026 claim	Status in this edition
The cut is Cramér–Rao saturation, $C \cdot \Omega = 1$	**Withdrawn.** The cut is the first time net forward Fisher–Rao travel reaches a half-turn, $a = \ell/2$, with $\Omega = 1/a$.
$\Omega_{Z2} = 1/\pi$ for bistable and $\Omega_{SO(2)} = 1/2\pi$ for rotational systems; ratio 2	**Withdrawn.** Ω = 1/(half-turn) for every carrier.
CV = 1/(2π) or 1/(4π) of inter-cut times; the breath at CV = 0.080	**Withdrawn.** Timing variability depends on each system's added dynamics, not on geometry.
Kernel $e^{C/\Omega}$: "+9.9 nats", "$\times 1.9 \times 10^4$", unitarity accounting	**Withdrawn.** Regeneration is a normalised, age-weighted Fréchet mean (§3).
"The rupture is screen-inducing"	**Withdrawn.** A reduced state exists at every instant of any bipartite pure state; nothing is special at the cut.
A 2:1 Arnold tongue "centred by topology"	**Withdrawn.** It depended on two different Ω.
Self-measurement overshoot (illustrated)	**Strengthened:** it is a theorem (§4.1).
Retensing: $C \geq S_A$ "robustly"	**Corrected:** a continuity bound whose constant exceeds 1 beyond small systems (§4.2).
The collective now through a synchronisation transition	**Kept, and reduced:** it is the Kuramoto transition (§4.4).
The boundary as a rate; an empty present; the far side as the settled past	**Kept,** as a philosophical position (§5).

2. The argument CRR responds to

Sandved-Smith and colleagues [1] start from the free-energy principle, on which to persist is to self-evidence: to maximise the evidence for one's generative model. Their argument has three steps:

1. Among the beliefs a self-evidencing system holds, one can never receive evidence: that it is bounded off from its environment.
2. Confirming separateness would require measuring the entanglement entropy across one's own boundary from inside, which a finite system cannot do. This is built on the quantum free-energy programme [3]. They formalise the belief in separation as a structural prior over the agent's quantum-reference-frame (QRF) deployments.
3. Interpretively, they read awakening as the embodied recognition of this impossibility.

An operational companion [2] implements these ideas as audited software surrogates, and its record states the central claim this way: a finite agent can use a boundary for

prediction without ever obtaining evidence that the boundary is real.

I take the impossibility result as correct and do not contest it.

3. CRR in its current form

All of the following follow the current CRR core (sections R1–R7).

- **Carrier.** A chosen statistical family carrying the Fisher–Rao metric.
- **Coherence.** C is accumulated Fisher–Rao arc in fixed units σ. The chord C* is the geodesic distance between the occasion's start and end, and the surplus is S = C − C*.
- **The cut.** On an identifiable circle of circumference ℓ, the cut is the first passage $t_{n+1} = \inf\{ t > t_n : s(t) - s(t_n) = a \}$, with $a = \ell/2$ and $\Omega = 1/a$, the same for every carrier. A monotone half-turn has S = 0, and in general S = 2B, where B is the backward travel within the occasion.
- **Physical laws.** They are constitutive additions, not consequences of CRR.
- **Regeneration.** The seed of the next occasion is an age-weighted Fréchet mean of completed contents: $m_{n+1} \in \operatorname{argmin}_y \Sigma_j \, q^{n-j} \, d(y, \Phi_j)^2$.
  - In flat coordinates this is the convex recursion $m_{n+1} = q \, m_n + (1 - q) \, \Phi_n$.
  - The weights are normalised. The map from a record of past contents to one seed is many-to-one, so it can neither amplify nor be inverted.
  - On a circle the mean can be non-unique, and a tie-break must then be declared.
- **The cost of the path.** Section R6 of the core gives a conditional lower bound on frictional dissipation over an occasion of duration T: $Q_n \, T_n \geq \zeta_{\min} \, \sigma^2 \, (C^*_n + S_n)^2$. Surplus therefore has a thermodynamic price at fixed duration. This replaces the June edition's vaguer "Ω must be paid in free energy", and it is existing thermodynamic-length mathematics, not a new law.
- **Ontology** (labelled as such in the core). Actual history accumulates; the present is an empty boundary that carries conditions inherited from that history; future content is not actual.

4. Four checks, preregistered and confirmed

Pipeline (the Phoenix protocol; study `BOUNDARY-RATE-CHECK-2026-10-27`):

1. **Development.** Seeds 0–2 were used for development and registered as seen.
2. **Pre-freeze correction.** A dry run on development seeds caught one of my own claims as wrong (§4.2, B6), and it was corrected before the freeze.
3. **Freeze.** The predictions, scorer and code were frozen (FREEZE.json sha256 `6d1a7669…94b9`).
4. **Anchor.** The freeze commit `b29a1c8` was pushed, and the freeze record was stamped with OpenTimestamps; the Bitcoin attestation is pending.
5. **Holdout.** Only then were the holdout instances drawn: random generator 2000, trajectory seeds 2000–2005 and Kuramoto seeds 2000–2002. All eight checks and

the harness check held: verdict RESULT. Two reruns were byte-identical.

6. **Ceiling.** No owner-signed tag preceded the run, so the ceiling is “PASS-0; weak anchor”.

4.1 Self-measurement overshoot is a theorem

The claim. Measure a state ρ in any orthonormal basis. The outcome probabilities are its diagonal in that basis, and they are majorised by its eigenvalues (Schur 1923; Horn 1954). So the Shannon entropy of the outcomes is at least the von Neumann entropy S(ρ), with equality exactly in the eigenbasis.

What it means here. An agent reading its own boundary entropy from any frame it can deploy overshoots, unless it already knows its eigenbasis, and finding that would require the full tomography a finite system lacks for its own boundary. The June edition presented this as an illustration on eight frames; it holds in general.

Holdout (B1). 500 random states and Haar-random bases at each of d = 2, 4, 8 and 16:

- the overshoot was ≥ 0 in every case;
- the mean overshoot was 0.25, 0.38, 0.44 and 0.47 nats respectively;
- equality held in the eigenbasis to 10^{-15}.

4.2 Retensing is continuity, and $C \geq S_A$ was a qubit property

The June claim. For a system A coupled to a bath E, starting from a product pure state, the accumulated Fisher–Rao arc C of A’s spectrum bounds the entanglement entropy $S_A = \tfrac{1}{2} I(A:E)$ “robustly”. The June edition read this as the forbidden spatial correlation being bounded by an accumulated temporal quantity.

What it actually is. Let p be A’s spectrum and $V(p) = \mathrm{Var}_p(-\ln p)$ its varentropy. Since $\Sigma\, dp_i = 0$, Cauchy–Schwarz gives

$$|dS| = |\Sigma\, (\ln p_i + S)\, dp_i| \leq \sqrt{V} \cdot \sqrt{(\Sigma\, dp_i^2/p_i)} = \sqrt{V}\, ds.$$

Integrating along the path:

$$\mathbf{S(t) \leq S(0) + \int \sqrt{V}\, dC.}$$

Equality holds when the spectrum flows along the entropy gradient.

The constant. Write κ(d) for the largest √V on d outcomes. Then:

- $\kappa(2) = 0.663$ nats, so for a qubit $C \geq S_A$ always holds. The June script used a qubit only.
- $\kappa(4) = 1.012$, $\kappa(8) = 1.333$ and $\kappa(32) = 1.949$, so beyond small systems the bound no longer implies $C \geq S_A$.
- **An exact construction.** The Fisher–Rao geodesic from a pure spectrum to the uniform one has length $2 \arccos(1/\sqrt{d})$. It is longer than ln d for $d \leq 13$ and

shorter for $d \geq 14$: at $d = 14$, $2.600 < 2.639$. So a spectrum can gain more entropy than the arc it travels.

- **The pre-freeze correction.** I first wrote this crossover as $d = 24$, from bounding the geodesic by π. The development dry run showed that to be wrong, and B6 was corrected before the freeze.

Holdout (B2–B6; system and bath both coupled through GUE Hamiltonians, 6 seeds per size, 241 time points):

- the continuity bound held at every point for $d_A = 2$, 4, 8 and 32;
- $C \geq S_A$ held at every point for $d_A = 2$ and 4;
- it failed on every seed at $d_A = 8$, by -0.03 to -0.05;
- it failed on every seed at $d_A = 32$, by -0.65 to -0.69, where S_A reached $1.38 \times C$ (Figure 1).

What this means. The forbidden quantity *is* bounded by an accumulated temporal one. But the bound is the ordinary continuity of entropy, and its constant depends on the system's size. It is not a CRR law, and it gives the agent no access to S_A: computing C requires A's spectrum, which is the very thing the agent cannot read across its own boundary.

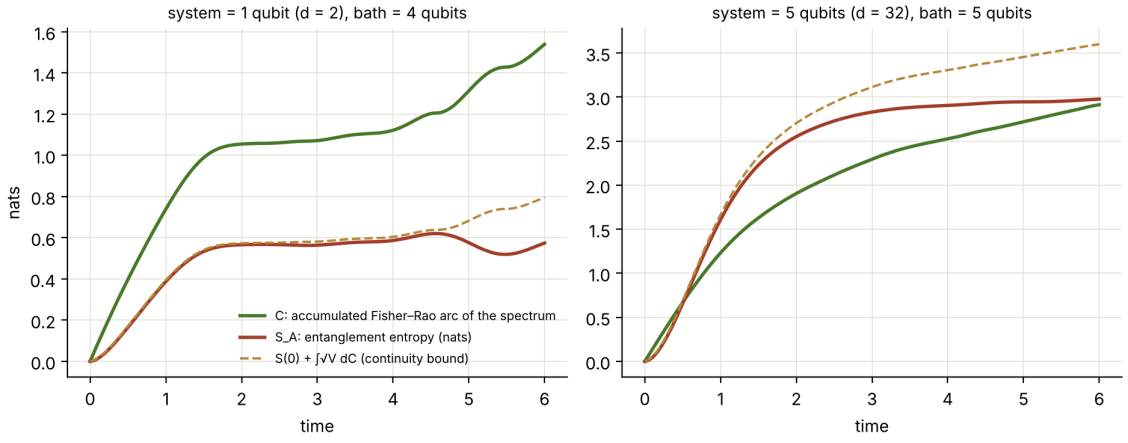


Figure 1. Accumulated Fisher–Rao arc C of the system's spectrum (green), the entanglement entropy S_A (red) and the continuity bound $S(0) + \int\sqrt{V}\, dC$ (gold, dashed), holdout seed 2000. Left: a one-qubit system, where C bounds S_A. Right: a five-qubit system, where S_A overtakes C while the continuity bound holds.

4.3 Regeneration neither amplifies nor inverts

The June edition's headline numbers for the kernel (+9.9 nats; $\times 1.9 \times 10^4$; a renormalised $\Delta H = -1.34$ nats) belonged to the withdrawn $e^{C/\Omega}$ form.

In the current form, regeneration is a normalised, age-weighted average. It stays within the convex hull of bounded contents and maps many different records to the same seed, so it cannot be unitary or even invertible. That is the kernel's whole "non-unitarity", and it is ordinary.

Any physical memory that implements it must pay for the record it overwrites. That is a Landauer-type cost, but its size depends on how the record is coarse-grained, and CRR does not fix that. Here the June edition's "debt" becomes something more precise and more modest: the surplus bound of §3, plus whatever the implementation of memory costs.

4.4 The collective now is the Kuramoto transition

The model. Take N CRR rotors, each a phase s_i on its own circle carrier, with the R4 law, no potential and no noise:

$ds_i/dt = \tau_i + K\, r \sin(\psi - s_i)$.

The tilts τ_i are Lorentzian with half-width γ, and $r\, e^{i\psi}$ is the mean field. This is exactly the Kuramoto model, for which $Kc = 2\gamma$ and $r = \sqrt{(1 - Kc/K)}$ above threshold, while $r = O(N^{-1/2})$ below it (Kuramoto 1975; Strogatz 2000).

Holdout (B7–B8):

- above threshold, at N = 4,000, the mean order parameter was within 0.006 of theory at K/Kc = 1.5, 2 and 3;
- below threshold, at K = Kc/2, r fell with N at a log–log slope of −0.375. The incoherent value is −½; finite run length makes the slope shallower, and the frozen band was [−0.65, −0.30] (Figure 2).

Because the tilts were deterministic Lorentzian quantiles, r above threshold was the same to four decimals on all three seeds; only the initial phases differed.

The June edition's 2:1 tongue "centred by topology" is withdrawn. It needed bistable and rotational systems to have different Ω.

What survives is a reading. Above a critical coupling, many finite systems share a common present, in the sense of phase coherence. Below it, each keeps a private present, and the shared one shrinks as $N^{-1/2}$. The transition is real and well known; calling it a "collective now" is interpretation.

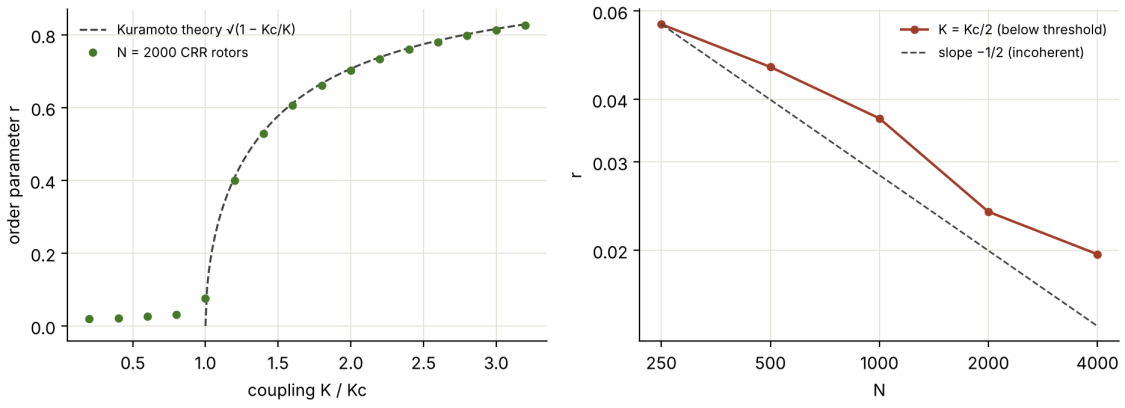


Figure 2. Left: order parameter of N = 2,000 coupled CRR rotors against coupling (green) and the Kuramoto theory (dashed). Right: below threshold (K = Kc/2), r falls with N towards the incoherent

slope −1/2 (holdout seed 2000).

5. The boundary as a rate (philosophical position)

Nothing in §4 proves the following; it is what the rebuilt mathematics leaves room for.

- **On the screen account,** at any instant a definite partition stands between self and environment, and the theorem forbids measuring across it. A determinate far side is kept behind the screen.
- **CRR does not begin with a partition.** What plays the part of the boundary is the advancing front of an occasion: its accumulated path toward the next cut. Every system has its own path and its own dynamics, so there is no single screen but many rates meeting.
- **The present carries no content of its own** (core, section VII). The content of an occasion is what has been accumulated, and the next occasion is seeded only from completed ones.
- **So the far side is not a hidden, co-present environment.** It is the settled past of other finite systems, taken up selectively. Each system's regeneration weights and averages what it inherits, and two systems meeting in the same now each enact the other through their own history.
- **This accepts the impossibility theorem rather than escaping it.** Computing even the retensing bound would require the agent's own spectrum (§4.2), and reading one's boundary entropy from inside overshoots (§4.1).
- **What changes is the picture of what lies beyond.** On the screen account it is an epistemic limit with a determinate far side. On this reading it is an ontological claim: no wall, only earlier.
- **Both are true by construction.** "The boundary is unevidenceable" is what the QRF formalism forbids. "Only the past has content" is what CRR's definitions make unavoidable. Neither is a discovery about the world. Each is what its formalism makes unavoidable, and saying so plainly is what lets the two be compared.

6. Resonances (interpretive only)

The June edition set out correspondences with contemplative and process traditions. I keep them only as resonances with §5, and remove the June claim that they "match" the withdrawn geometry, such as the bistable arc as half of the rotational circle.

- **Nāgārjuna** (*Mūlamadhyamakakārikā* 24.14 and 24.18 [5]). For one for whom emptiness is possible, everything is possible, and dependent arising is the middle way. Emptiness is enabling, not privative, as a present without content of its own lets every past arrive.
- **The Lurianic *tzimtzum*** (contraction opening a vacated space [8]). It names a vacated centre.
- **Zhuangzi's "pivot of the Way"** [6]. A centre that responds without end because it is fixed nowhere.

- **Ibn ’Arabī’s distinction** between the child and the owner of the moment [7]. Agency over what is taken up from the past, not over the rate at which occasions come.
- **Doctor and colleagues’ care and cognitive light cone** [4]. Here they read naturally as the reach of what a system can hold as a goal, and as the distance still to travel before its next cut.

These correspondences carry no formal weight.

7. Limits

- **All results are checks under chosen models.** None is contact with empirical data.
- **C here is the arc of A’s spectrum, not of its full state.** Eigenvector motion would only lengthen it, so the continuity bound and the violations both stand.
- **The model classes are narrow.** The Hamiltonians are one class (GUE), and the Kuramoto rotors are noise-free with a deterministic sample of tilts.
- **The operator is not independent.** The same model that wrote the study ran it. Ceiling “PASS-0; weak anchor”; any ledger entry awaits the owner’s review.
- **One source was confirmed indirectly.** I confirmed reference [1] through the citing record of [2], not from the preprint page, which did not render for me.

8. Reproduction

Python 3.13, numpy 2.5.3 and scipy 1.18.1 (matplotlib for the figures). From the study directory:

```
sh run.sh RUNS_DIR        # all holdout checks: outcomes.json and verdict.json (about 10 minutes)
python3 -I controls.py    # controls scored before the freeze
python3 figures.py        # Figures 1 and 2
```

The full code is listed in the Appendix and lives in the Phoenix repository under `Phoenix_Egg/studies/BOUNDARY-RATE-CHECK-2026-10-27/`. The receipts are under `Phoenix_Egg/receipts/BOUNDARY-RATE-CHECK-2026-10-27/`.

References

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Appendix: complete code

Each file is reproduced verbatim from the frozen study directory.

boundary_model.py

```
        """Model code for BOUNDARY-RATE-CHECK-2026-10-27 (frozen). Rebuilds the formal checks of
        "The Boundary is a Rate" (June 2026) under the current CRR form (CORE/MATHEMATICS.md R1-
R4).

        Three parts, each a known result exercised on fresh random instances:
         1. Self-measurement overshoot (Schur-Horn): the Shannon entropy of measurement outcomes of
rho in any basis
            is >= the von Neumann entropy S(rho), with equality in the eigenbasis.
         2. Retensing as continuity: for a spectrum p(t) moving with Fisher-Rao speed, |dS/ds| <=
sqrt(V(p)), V the
            varentropy (variance of -ln p). Hence S(t) <= S(0) + int sqrt(V) dC, with C the
accumulated Fisher-Rao arc of
            the spectrum (CRR coherence at sigma = 1). sup sqrt(V) is 0.663 nats for a qubit but
exceeds 1 from d = 4,
            so the June claim C >= S_A is a qubit property, not a law.
         3. The collective now: N CRR rotors on circle carriers with the R4 law (no potential, no
noise),
            ds_i/dt = tau_i + K r sin(psi - s_i), tau_i Lorentzian (half-width gamma): the Kuramoto
model.
            r(K) -> sqrt(1 - Kc/K), Kc = 2 gamma (Kuramoto 1975; exact for the Lorentzian).
        """
        import numpy as np
        from numpy.linalg import eigh, eigvalsh

        # ---------- 1. self-measurement overshoot ----------
        def haar_unitary(d, rng):
            z = (rng.standard_normal((d, d)) + 1j * rng.standard_normal((d, d))) / np.sqrt(2)
            q, r = np.linalg.qr(z); ph = np.diag(r) / np.abs(np.diag(r)); return q * ph

        def random_state(d, rng):
            g = rng.standard_normal((d, d)) + 1j * rng.standard_normal((d, d)); rho = g @
g.conj().T; return rho / np.trace(rho).real

        def H(p):
            p = np.clip(np.real(p), 0, None); p = p / p.sum(); nz = p > 0; return float(-(p[nz] *
np.log(p[nz])).sum())

        def overshoot(d, rng):
            rho = random_state(d, rng); U = haar_unitary(d, rng)
            S = H(eigvalsh(rho)); Hb = H(np.diag(U.conj().T @ rho @ U)); w, V = eigh(rho); He =
H(np.diag(V.conj().T @ rho @ V))
            return Hb - S, abs(He - S)

        # ---------- 2. retensing ----------
        def gue(d, rng):
            A = rng.standard_normal((d, d)) + 1j * rng.standard_normal((d, d)); return (A +
A.conj().T) / (2 * np.sqrt(2 * d))

        def trajectory(nA, nB, seed, g=1.0, tmax=6.0, npts=241):
            """System of nA qubits, bath of nB qubits, H = H_A + H_B + g H_AB (all GUE), product
initial state."""
            rng = np.random.default_rng(seed); dA, dB = 2 ** nA, 2 ** nB
            Hm = np.kron(gue(dA, rng), np.eye(dB)) + np.kron(np.eye(dA), gue(dB, rng)) + g * gue(dA
* dB, rng)
            E, V = eigh(Hm)
            a = rng.standard_normal(dA) + 1j * rng.standard_normal(dA); a /= np.linalg.norm(a)
            b = rng.standard_normal(dB) + 1j * rng.standard_normal(dB); b /= np.linalg.norm(b)
            c0 = V.conj().T @ np.kron(a, b); ts = np.linspace(0, tmax, npts); P = []; norm_err =
0.0
```

```
            for t in ts:
                psi = V @ (np.exp(-1j * E * t) * c0); norm_err = max(norm_err, abs(np.vdot(psi,
psi).real - 1))
                M = psi.reshape(dA, dB); ev = np.clip(eigvalsh(M @ M.conj().T).real, 1e-300, None);
P.append(np.sort(ev / ev.sum())[::-1])
            P = np.array(P); S = -(P * np.log(P)).sum(1)
            step = 2 * np.arccos(np.clip(np.sqrt(P[1:] * P[:-1]).sum(1), -1, 1))
            C = np.concatenate([[0.0], np.cumsum(step)])
            l = np.log(P); m = (P * l).sum(1, keepdims=True); sV = np.sqrt((P * (l - m) **
2).sum(1))
            # sup of sqrt(V) along each chord, taken at the endpoints plus a midpoint spectrum
(conservative for small steps)
            mid = 0.5 * (P[1:] + P[:-1]); lm = np.log(mid); mm = (mid * lm).sum(1, keepdims=True);
sVm = np.sqrt((mid * (lm - mm) ** 2).sum(1))
            sup = np.maximum(np.maximum(sV[1:], sV[:-1]), sVm)
            bound = S[0] + np.concatenate([[0.0], np.cumsum(step * sup)])
            return dict(ts=ts, S=S, C=C, bound=bound, norm_err=norm_err)

        def max_root_varentropy(d, starts=60, seed=0):
            from scipy.optimize import minimize
            rng = np.random.default_rng(seed); best = 0.0
            def f(x):
                p = np.exp(x - x.max()); p /= p.sum(); l = np.log(np.clip(p, 1e-300, None)); m = (p
* l).sum(); return -np.sqrt((p * (l - m) ** 2).sum())
            for _ in range(starts):
                r = minimize(f, rng.normal(size=d) * 3, method="Nelder-Mead",
options=dict(maxiter=20000, xatol=1e-10, fatol=1e-12)); best = max(best, -r.fun)
            return best

        def geodesic_counterexample(d):
            """Fisher-Rao geodesic length from a pure spectrum to the uniform one vs the entropy
gained (ln d)."""
            return 2 * np.arccos(1 / np.sqrt(d)), np.log(d)

        # ---------- 3. collective now ----------
        def kuramoto_r(N, K, gamma, seed, T=300.0, dt=0.05, burn=150.0):
            """Mean order parameter of N CRR rotors (R4 law, no potential/noise) with deterministic
Lorentzian tilts."""
            rng = np.random.default_rng(seed)
            q = (np.arange(1, N + 1) / (N + 1)) - 0.5; tau = gamma * np.tan(np.pi * q)
            s = rng.uniform(0, 2 * np.pi, N); rs = []; nsteps = int(T / dt); nb = int(burn / dt)
            for k in range(nsteps):
                z = np.exp(1j * s).mean(); r, psi = abs(z), np.angle(z)
                s = s + dt * (tau + K * r * np.sin(psi - s))
                if k >= nb: rs.append(r)
            return float(np.mean(rs))
```

run_checks.py

```
        """Holdout runner for BOUNDARY-RATE-CHECK-2026-10-27. Usage: python3 -I run_checks.py
OUT.json
        Holdout seeds: overshoot draws from default_rng(2000); trajectories seeds 2000-2005;
Kuramoto seeds 2000-2002."""
        import json, sys, pathlib
        sys.path.insert(0, str(pathlib.Path(__file__).resolve().parent))
        import numpy as np
        from boundary_model import overshoot, trajectory, kuramoto_r, max_root_varentropy,
geodesic_counterexample

        def rnd(x):
            if isinstance(x, float): return round(x, 6)
            if isinstance(x, list): return [rnd(v) for v in x]
            if isinstance(x, dict): return {k: rnd(v) for k, v in x.items()}
            return x

        out = {}
        rng = np.random.default_rng(2000)
```

```
        out["overshoot"] = {}
        for d in (2, 4, 8, 16):
            o = [overshoot(d, rng) for _ in range(500)]
            out["overshoot"][str(d)] = dict(n=500, min_gap=float(min(x[0] for x in o)),
mean_gap=float(np.mean([x[0] for x in o])),
                                            max_eigenbasis_err=float(max(x[1] for x in o)))
        out["retensing"] = {}
        for nA, nB in [(1, 4), (2, 4), (3, 5), (5, 5)]:
            rows = []
            for seed in range(2000, 2006):
                r = trajectory(nA, nB, seed)
                rows.append(dict(seed=seed, min_C_minus_S=float(np.min(r["C"] - r["S"])),
min_bound_minus_S=float(np.min(r["bound"] - r["S"])),
                                 max_S_over_C=float(np.max(r["S"][1:] / r["C"][1:])),
final_S=float(r["S"][-1]), final_C=float(r["C"][-1]), norm_err=float(r["norm_err"])))
            out["retensing"][str(2 ** nA)] = rows
        out["kappa"] = {str(d): max_root_varentropy(d) for d in (2, 4, 8, 32)}
        out["geodesic"] = {str(d): dict(zip(("C_geodesic", "S_uniform"), map(float,
geodesic_counterexample(d)))) for d in (13, 14, 32, 64)}
        g = 0.5; Kc = 2 * g
        out["kuramoto_above"] = {str(m): dict(theory=float(np.sqrt(1 - 1 / m)), r=[kuramoto_r(4000,
m * Kc, g, s) for s in range(2000, 2003)]) for m in (1.5, 2.0, 3.0)}
        Ns = [250, 500, 1000, 2000, 4000]
        rs = [float(np.mean([kuramoto_r(N, 0.5 * Kc, g, s) for s in range(2000, 2003)])) for N in
Ns]
        out["kuramoto_below"] = dict(N=Ns, r=rs, slope=float(np.polyfit(np.log(Ns), np.log(rs), 1)
[0]))
        pathlib.Path(sys.argv[1]).write_text(json.dumps(rnd(out), indent=1, sort_keys=True) + "\n")
```

score.py

```
        """Frozen scorer for BOUNDARY-RATE-CHECK-2026-10-27. Usage: python3 -I score.py
outcomes.json > verdict.json"""
        import json, sys
        o = json.load(open(sys.argv[1])); R = o["retensing"]
        checks = {
         "V0_harness_unitarity": all(r["norm_err"] < 1e-10 for d in R for r in R[d]),
         "B1_overshoot_schur_horn": all(v["min_gap"] >= -1e-10 and v["mean_gap"] > 0 and
v["max_eigenbasis_err"] < 1e-10 for v in o["overshoot"].values()),
         "B2_continuity_bound_holds": all(r["min_bound_minus_S"] >= -1e-9 for d in R for r in
R[d]),
         "B3_qubit_C_bounds_S": all(r["min_C_minus_S"] >= -1e-9 for r in R["2"]),
         "B4_C_ge_S_fails_at_dA_32": all(r["min_C_minus_S"] < -0.3 for r in R["32"]),
         "B5_kappa_crosses_one": o["kappa"]["2"] < 1 < o["kappa"]["4"],
         "B6_geodesic_counterexample": o["geodesic"]["14"]["C_geodesic"] < o["geodesic"]["14"]
["S_uniform"] and o["geodesic"]["13"]["C_geodesic"] > o["geodesic"]["13"]["S_uniform"],
         "B7_kuramoto_above_Kc": all(abs(sum(v["r"]) / len(v["r"]) - v["theory"]) <= 0.03 for v in
o["kuramoto_above"].values()),
         "B8_kuramoto_below_Kc_finite_size": -0.65 <= o["kuramoto_below"]["slope"] <= -0.30,
        }
        verdict = "VOID" if not checks["V0_harness_unitarity"] else ("RESULT" if
all(checks.values()) else "FAIL")
        json.dump({"study_id": "BOUNDARY-RATE-CHECK-2026-10-27", "checks": checks, "verdict":
verdict}, sys.stdout, indent=1, sort_keys=True); print()
```

controls.py

```
        """Controls for BOUNDARY-RATE-CHECK-2026-10-27, scored before the freeze.
        pos_bound_is_tight: along an entropy-gradient flow of the spectrum (d = 8) the continuity
bound is nearly attained,
          so it is not vacuous (ratio of entropy gained to int sqrt(V) dC >= 0.99).
        neg_geodesic_breaks_C_ge_S: along the Fisher-Rao geodesic from a pure to the uniform
spectrum at d = 32, C < S at the end
          (the scorer's B4 logic must be able to register a violation).
        null_kuramoto_K0: with no coupling r ~ 0.886/sqrt(N) (incoherent), r*sqrt(N) in [0.5, 1.5]
```

```
at N = 2000.
        Usage: python3 -I controls.py > controls.json"""
        import json, sys, pathlib
        sys.path.insert(0, str(pathlib.Path(__file__).resolve().parent))
        import numpy as np
        from boundary_model import kuramoto_r
        d = 8; p = np.array([0.93] + [0.01] * 7); S0 = -(p * np.log(p)).sum(); integral = 0.0
        for _ in range(20000):                        # gradient flow of S in the Fisher metric:
dp_i = eps p_i (-ln p_i - S)
            l = np.log(p); S = -(p * l).sum(); dp = 1e-4 * p * (-l - S); q = p + dp; q /= q.sum()
            ds = 2 * np.arccos(min(1.0, np.sqrt(p * q).sum())); lm = np.log(p); m = (p * lm).sum();
integral += ds * np.sqrt((p * (lm - m) ** 2).sum()); p = q
        gain = -(p * np.log(p)).sum() - S0
        x = np.linspace(0, 1, 2001); d2 = 32; e = np.zeros(d2); e[0] = 1; u = np.ones(d2) / d2   #
geodesic on the sphere of sqrt-probabilities
        th = np.arccos(np.sqrt(e * u).sum()); path = [((np.sin((1 - t) * th) * np.sqrt(e) +
np.sin(t * th) * np.sqrt(u)) / np.sin(th)) ** 2 for t in x]
        C_end = sum(2 * np.arccos(min(1.0, np.sqrt(a * b).sum())) for a, b in zip(path[:-1],
path[1:])); S_end = np.log(d2)
        r0 = kuramoto_r(2000, 0.0, 0.5, 7000)
        out = {"pos_bound_is_tight": {"ratio": round(gain / integral, 4), "pass": bool(gain /
integral >= 0.99)},
               "neg_geodesic_breaks_C_ge_S": {"C_end": round(C_end, 4), "S_end": round(S_end, 4),
"violation_registered": bool(C_end < S_end)},
               "null_kuramoto_K0": {"r_sqrtN": round(r0 * np.sqrt(2000), 3), "pass": bool(0.5 <= r0
* np.sqrt(2000) <= 1.5)}}
        out["controls_behave"] = out["pos_bound_is_tight"]["pass"] and
out["neg_geodesic_breaks_C_ge_S"]["violation_registered"] and out["null_kuramoto_K0"]["pass"]
        print(json.dumps(out, indent=1, sort_keys=True))
```

run.sh

```
        #!/bin/sh
        # usage: sh run.sh RUNS_DIR   (after the freeze and anchor; holdout seeds 2000+ are drawn
for the first time)
        set -e
        H="$(cd "$(dirname "$0")" && pwd)"; R="$1"; mkdir -p "$R"
        export PYTHONDONTWRITEBYTECODE=1
        python3 -I "$H/run_checks.py" "$R/outcomes.json"
        python3 -I "$H/score.py" "$R/outcomes.json" > "$R/verdict.json"
```

figures.py

```
        """Figures for the rebuilt "The Boundary is a Rate" (holdout seed 2000 / Kuramoto seed
2000, identical to the frozen run).
        Usage: python3 figures.py   (writes ../fig_retensing.png and ../fig_collective.png)"""
        import numpy as np
        import matplotlib; matplotlib.use("Agg")
        import matplotlib.pyplot as plt
        from boundary_model import trajectory, kuramoto_r

        G, GOLD, RUST, INK = "#4A7A2C", "#B8893A", "#A3402B", "#444444"
        fig, axes = plt.subplots(1, 2, figsize=(9.6, 3.9), dpi=200)
        for ax, (nA, nB, lab) in zip(axes, [(1, 4, "system = 1 qubit (d = 2), bath = 4 qubits"),
(5, 5, "system = 5 qubits (d = 32), bath = 5 qubits")]):
            r = trajectory(nA, nB, 2000)
            ax.plot(r["ts"], r["C"], color=G, lw=2, label="C: accumulated Fisher–Rao arc of the
spectrum")
            ax.plot(r["ts"], r["S"], color=RUST, lw=2, label="S_A: entanglement entropy (nats)")
            ax.plot(r["ts"], r["bound"], color=GOLD, lw=1.4, ls="--", label="S(0) + ∫√V dC
(continuity bound)")
            ax.set_title(lab, fontsize=9.5); ax.set_xlabel("time")
            for sp in ("top", "right"): ax.spines[sp].set_visible(False)
            ax.grid(color="#e6e2d8", lw=0.6)
        axes[0].set_ylabel("nats"); axes[0].legend(frameon=False, fontsize=7.5, loc="lower right")
```

```
        fig.tight_layout(); fig.savefig("../fig_retensing.png", facecolor="white")

        fig, (a1, a2) = plt.subplots(1, 2, figsize=(9.6, 3.6), dpi=200); g = 0.5; Kc = 2 * g
        Ks = np.linspace(0.2, 3.2, 16) * Kc; rs = [kuramoto_r(2000, K, g, 2000) for K in Ks]
        kk = np.linspace(Kc, 3.2 * Kc, 200)
        a1.plot(kk / Kc, np.sqrt(1 - Kc / kk), color=INK, lw=1.2, ls="--", label="Kuramoto theory
√(1 − Kc/K)")
        a1.plot(Ks / Kc, rs, "o", color=G, ms=4, label="N = 2000 CRR rotors")
        a1.set_xlabel("coupling K / Kc"); a1.set_ylabel("order parameter r");
a1.legend(frameon=False, fontsize=8)
        Ns = [250, 500, 1000, 2000, 4000]; rb = [kuramoto_r(N, 0.5 * Kc, g, 2000) for N in Ns]
        a2.loglog(Ns, rb, "o-", color=RUST, ms=4, label="K = Kc/2 (below threshold)")
        a2.loglog(Ns, rb[0] * (np.array(Ns) / Ns[0]) ** -0.5, color=INK, lw=1, ls="--",
label="slope −1/2 (incoherent)")
        a2.minorticks_off(); a2.set_xticks(Ns); a2.set_xticklabels([str(n) for n in Ns]);
a2.set_yticks([0.02, 0.03, 0.04, 0.06]); a2.set_yticklabels(["0.02", "0.03", "0.04", "0.06"])
        a2.set_xlabel("N"); a2.set_ylabel("r"); a2.legend(frameon=False, fontsize=8)
        for ax in (a1, a2):
            for sp in ("top", "right"): ax.spines[sp].set_visible(False)
            ax.grid(color="#e6e2d8", lw=0.6)
        fig.tight_layout(); fig.savefig("../fig_collective.png", facecolor="white")
```